

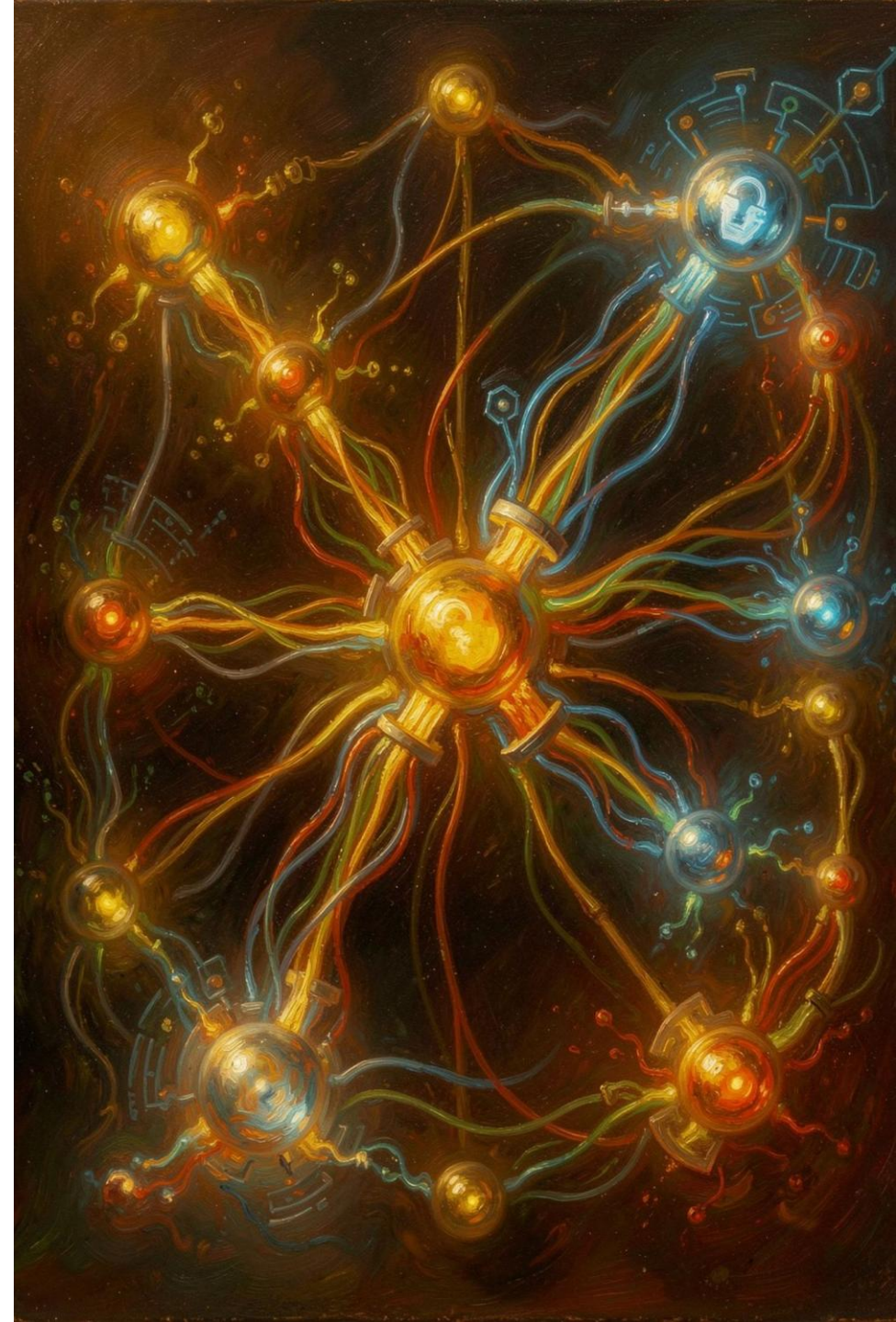
MILESTONE 4

Risk Intelligence & Executive Analytics

Advanced Threat Incident Analytics and Security Visualization Framework
— transforming raw incident data into actionable executive intelligence.

Submitted by:
Rohith
Srinivas
Samiksha

Guide:
Mr. Sugandhar



What Milestone 4 Delivers

Four interconnected analytical pillars — from risk classification to automated decision support.



Risk Intelligence

- Four-level risk classification
- Threat Severity Index (TSI) analysis
- Risk distribution by location and attack type



Predictive Analytics

- Incident forecasting
- Anomaly detection
- Threat severity trend analysis



Executive Analytics

- Executive KPIs and financial impact
- Response performance metrics
- SLA monitoring



Decision Support

- Automated risk alerts
- Management recommendations

Four-Level Security Risk Classification

The original three-level risk model was extended to four tiers using the Threat Severity Index (TSI), enabling more granular prioritization. Risk thresholds are project-defined analytical thresholds.

TSI Score	Risk Level	Incidents	Share
< 0.40	Low	2,091	20.91%
0.40 – 0.649	Medium	5,939	59.39%
0.65 – 0.849	High	1,906	19.06%
≥ 0.85	Critical	64	0.64%

 **Key Finding:** 19.70% of all incidents are classified as High or Critical — requiring prioritized investigation and response.

Where Is Security Risk Concentrated?

The Security Risk Assessment dashboard analyzes risk across geographic, attack, and financial dimensions to guide resource allocation.



Geographic Risk

Identifies states with higher concentrations of High/Critical incidents, enabling targeted investigation and resource deployment.



Financial Exposure

Compares financial losses across all risk levels, supporting prioritization based on potential business impact.



Attack-Type Risk

Identifies attack categories driving High/Critical incidents, helping prioritize security controls and investigation efforts.

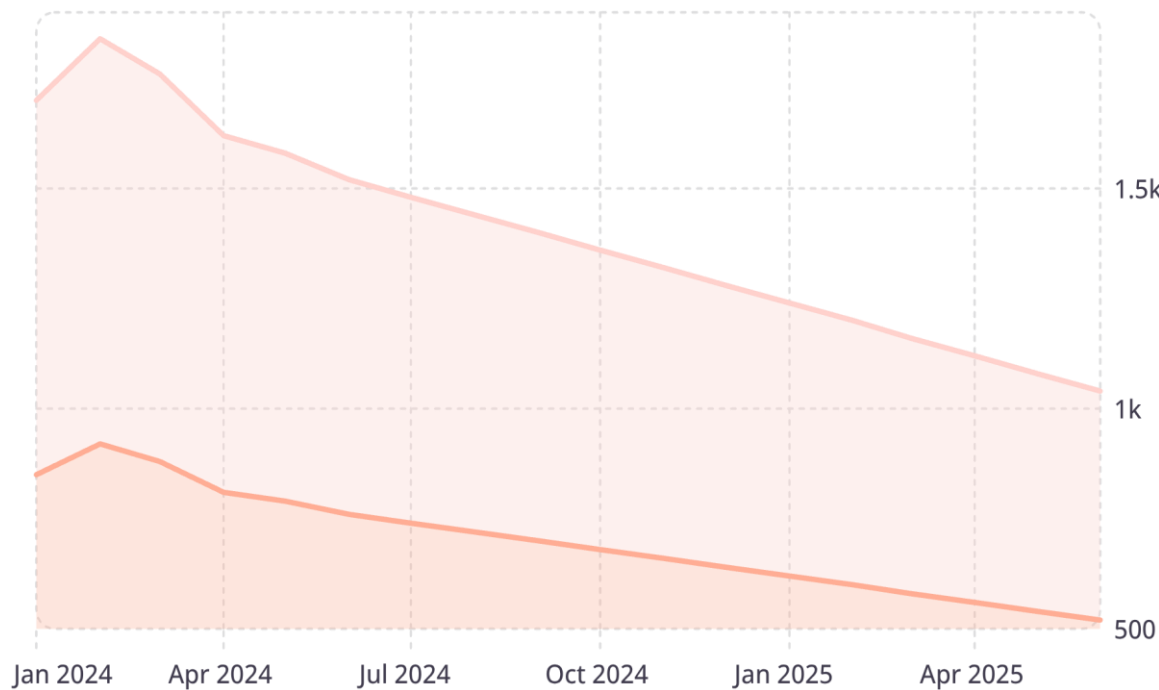


Risk Distribution

State-level risk matrix provides a detailed view of Low, Medium, High, and Critical incidents across the enterprise.

Cybersecurity Incident Forecast — Next 3 Months

Forecasted Incidents Actual Incidents



Forecast Configuration

- **Horizon:** Next 3 months
- **Confidence Interval:** 95%
- **Seasonality:** Auto-detected
- **Input:** Monthly incident counts

Business Value

The forecast indicates a **declining incident-volume trend** over the next three months, though the confidence interval is relatively wide — indicating uncertainty in predicted values.

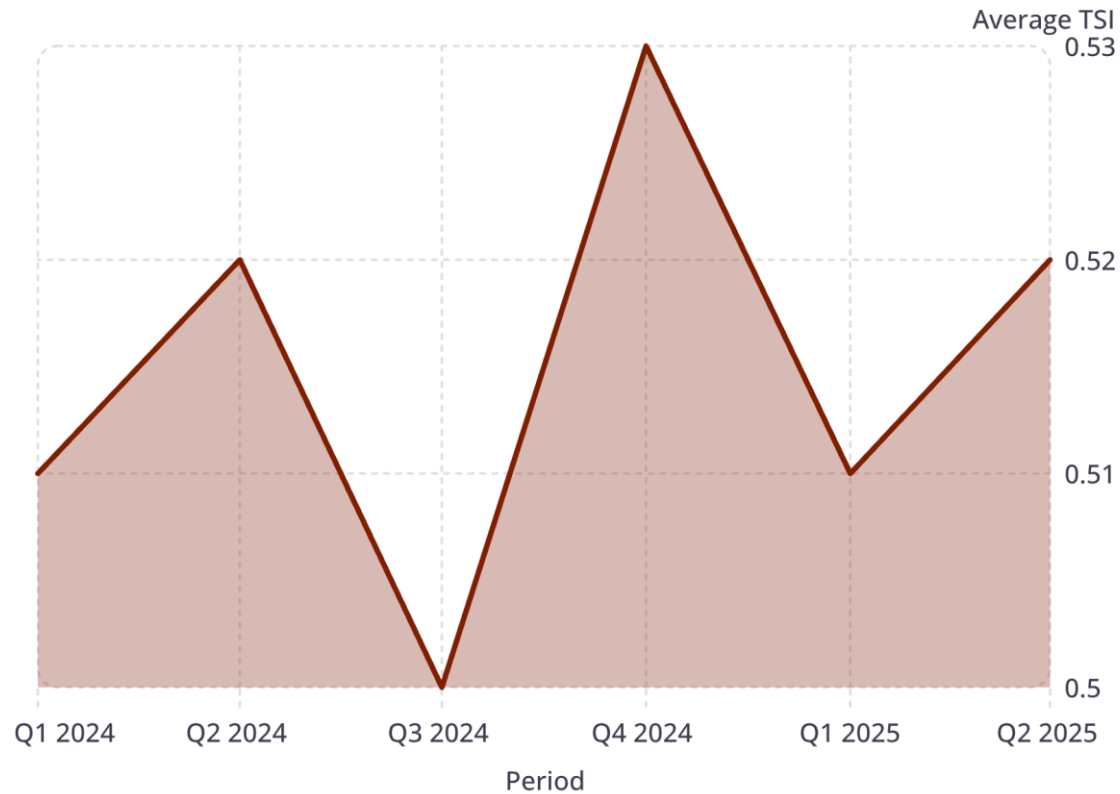
→ Security resource planning^{Month}

→ Operational preparedness

→ Incident-response workload planning

→ Proactive decision-making

Incident Anomaly Detection & Average TSI Trend



Anomaly Detection Findings

Power BI anomaly detection identified two unusual periods in incident volume:

Average TSI Trend

Average TSI remained between **0.50 and 0.53** with periodic fluctuations. No strong long-term increase or decrease was identified.

Key Insight: Incident volume contains isolated unusual periods, while overall threat severity remains **relatively stable**.

Early 2024

Unusually high incident activity detected.

Early 2025

Unusually low incident activity detected.

Executive Security Posture at a Glance

10K

Total Incidents

0.52

Average TSI

~2K

High/Critical Risks

~₹50B

Total Financial Loss

153m

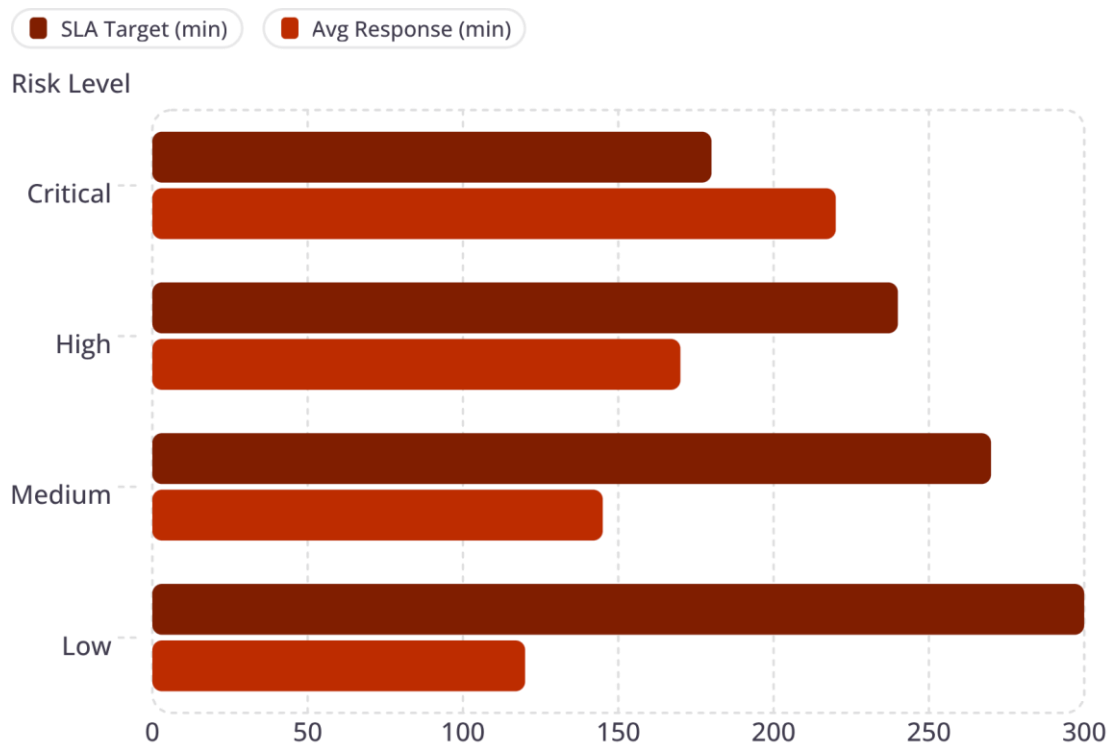
Avg Response Time

88%

SLA Compliance

Management Focus: 19.70% of incidents are High/Critical while average response time stands at 153.33 minutes. Critical incidents have the longest average response time. Prioritize High/Critical incidents while improving response performance for Critical-severity events.

Risk-Based SLA Monitoring



SLA Targets by Risk Level

Response expectations are assigned by risk level — not a single uniform target — ensuring critical incidents receive prioritized attention.

Key Finding: Response time increases with risk level. Critical incidents average **~220 minutes** — exceeding their 180-minute SLA target. Overall SLA compliance: **88.29%**.

Critical ≤ 180 min

High ≤ 240 min

Medium ≤ 270 min

Low ≤ 300 min

Automated Security Risk Alerts

Three alert conditions provide real-time, actionable risk signals for management. All three are currently triggered.

🔴 Risk Exposure Alert

Condition: High + Critical incidents > 15% of total

Current: 19.70%

Status: ALERT

🔴 SLA Breach Alert

Condition: SLA Compliance < 90%

Current: 88.29%

Status: ALERT

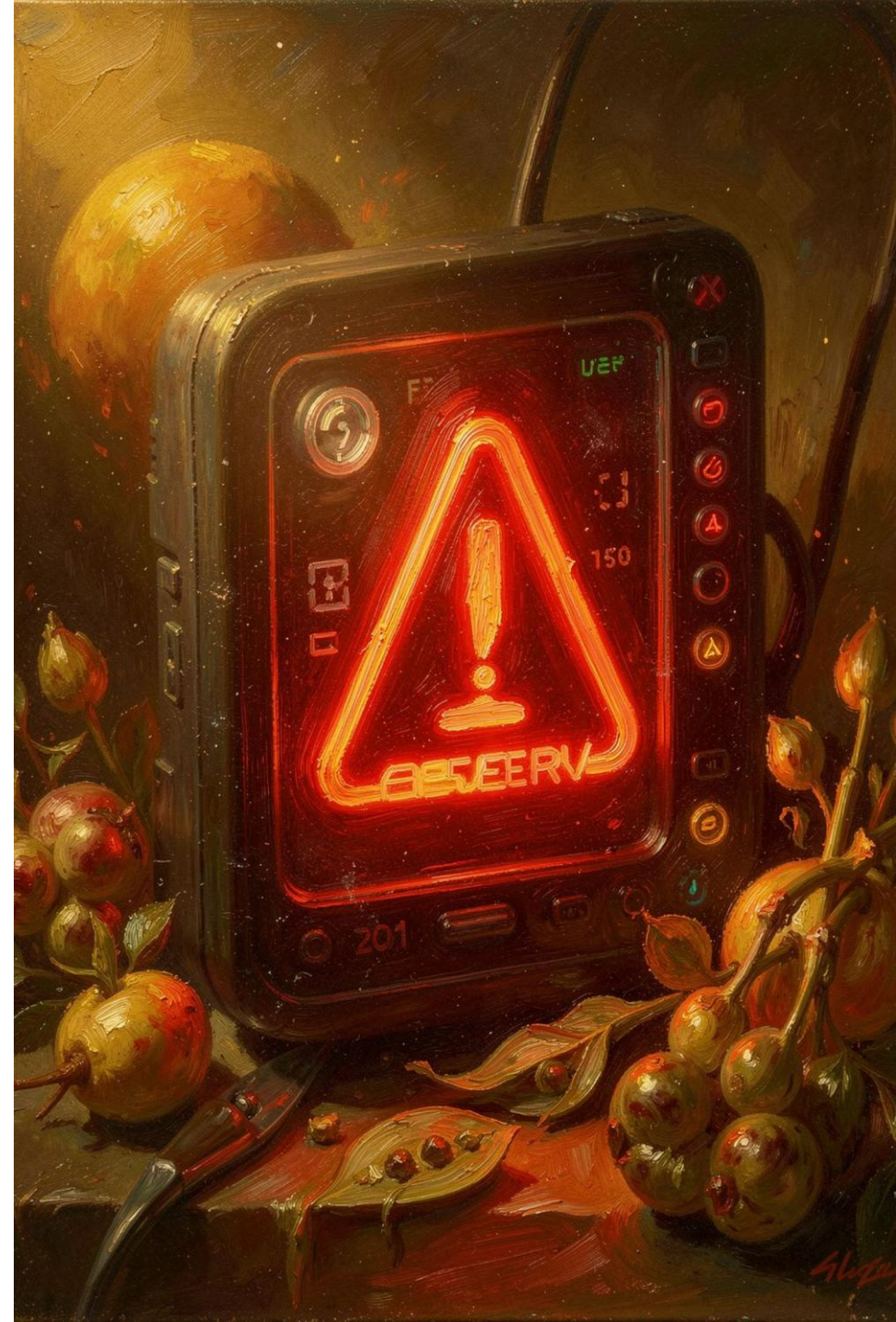
🔴 Critical Response Alert

Condition: Critical avg response time > 180 min

Current: ~220 min

Status: ALERT

⚠️ **All three alert conditions are currently triggered**, highlighting significant risk exposure and response-performance gaps requiring immediate management attention.



Actionable Priorities for Security Leadership

1 Prioritize High/Critical Incidents

Allocate additional security resources to the most affected states and attack types. Focus investigation efforts on the 19.70% of incidents classified as High or Critical.

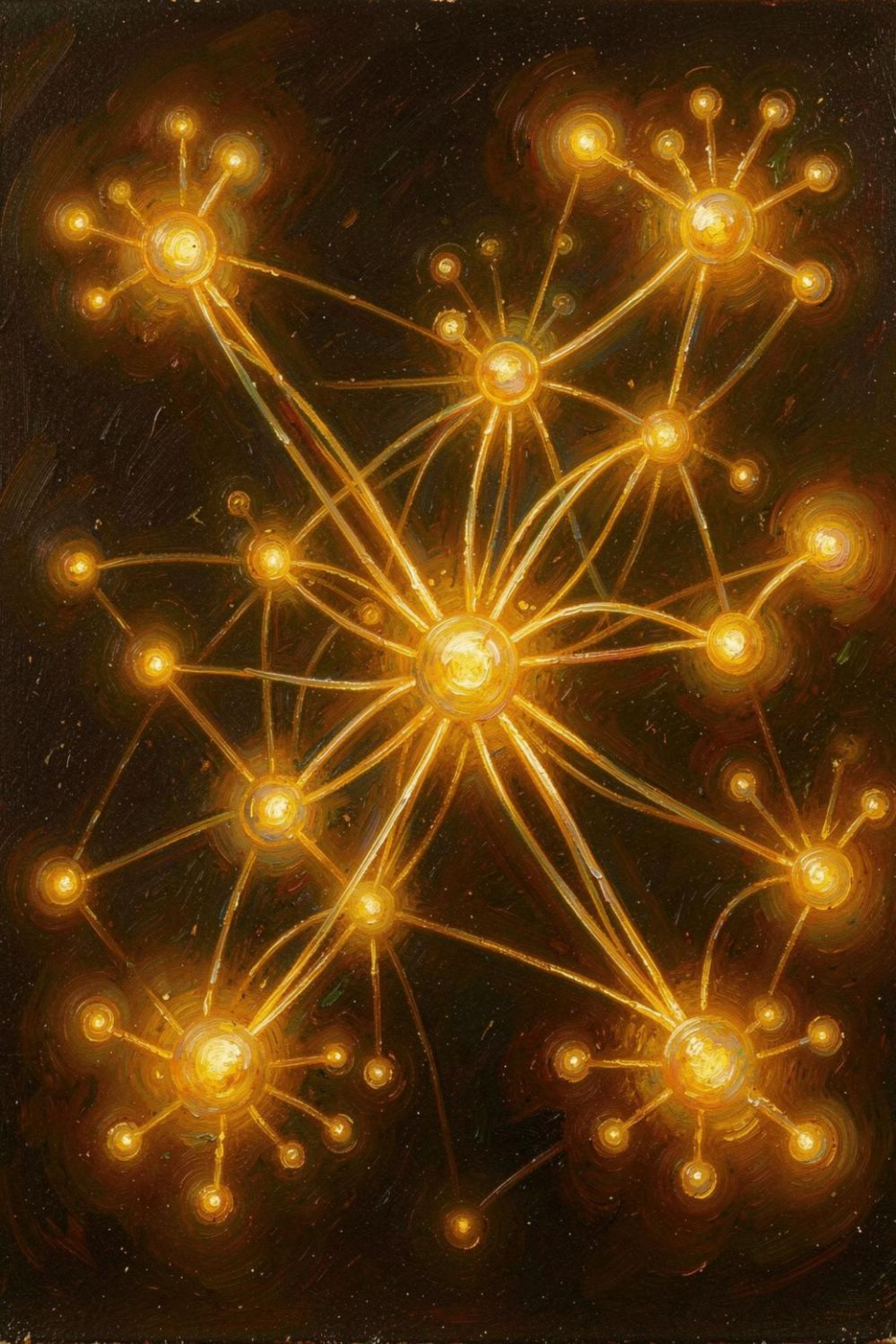
2 Improve Response Performance

Review incident-response workflows, escalation procedures, and staffing to reduce response delays and improve SLA compliance from 88.29% toward the 90% target.

3 Strengthen Critical Escalation

Immediately escalate Critical incidents and prioritize dedicated response resources to reduce time-to-containment from ~220 minutes toward the 180-minute SLA target.

Overall Objective: Improve incident prioritization, response efficiency, and resource allocation based on quantified security risk.



MILESTONE 4 — CONCLUSION

An Integrated Framework for Security Intelligence

Milestone 4 transforms cybersecurity incident data into actionable executive intelligence — enabling proactive, risk-informed decision-making.

01 — Risk Intelligence

Four-level classification, geographic and attack-type analysis, financial exposure

02 — Predictive Analytics

3-month forecasting, anomaly detection, TSI trend analysis

03 — Executive Analytics

Security KPIs, risk-based SLA monitoring, response and financial performance

04 — Decision Support

Automated risk alerts, management recommendations, proactive prioritization